

Residual-Basic Charges and Simultaneous Horizon First Laws on the Mannheim–Kazanas Family

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Working draft, 26 July 2026

Abstract

Static spherical solutions of four-dimensional Weyl gravity possess a residual local $\text{Diff} \times \text{Weyl}$ freedom that changes the normalization of the chart Killing vector. We show, by exact covariant-phase-space calculation, that this freedom resolves an apparent nonintegrability of the static Iyer–Wald charge. On the Mannheim–Kazanas component of a Laurent-complete Bach-flat family, the bare charge associated with ∂_t is not closed. Requiring the field-dependent generator and its charge to descend to the residual quotient forces

$$\chi = uf(J)\partial_t, \quad u = \beta(2 - 3\beta\gamma),$$

where J is the single continuous residual invariant. For the canonical choice $f = 1$, the corrected charge is exact, with residual-basic Hamiltonian

$$H = -16\pi\alpha\beta^2(9\beta^2\gamma^2k - \beta\gamma^3 - 12\beta\gamma k + \gamma^2 + 4k).$$

At every locally labelled simple root r_h of the metric function, the Wald entropy and signed horizon temperature

$$S_h = 64\pi^2\alpha\beta\frac{2 - 3\beta\gamma + \gamma r_h}{r_h}, \quad T_h = \frac{uB'(r_h)}{4\pi}$$

obey $dH = T_h dS_h$. The same one-form dH therefore gives simultaneous first laws at all simple horizons, including an exact non-Einstein three-horizon example. At linear charge level the result is unchanged by arbitrary time-dependent spherical Weyl rescalings and diffeomorphisms: those directions carry zero charge, zero entropy variation, and zero corrected presymplectic pairing with the static parameter modes.

The theorem is deliberately scoped. The background classification is complete only in the declared Laurent class, the charge theorem is exact on the static parameter family and in the linear spherical gauge sector, and no preferred physical mass, second-order physical-process first law, radiative flux law, stability statement, or quantum claim is inferred.

AI authorship and computational accountability

GPT-5.6.sol developed the derivations, symbolic implementations, verification architecture, claim audit, and manuscript. Asger Alstrup Palm commissioned and orchestrated the programme and is the corresponding human contact. The project is also an experiment in whether a non-specialist human, using AI systems as research collaborators, can organize falsifiable computer-assisted mathematical physics. Every promoted identity in this paper is bound to a machine-readable exact certificate and to a separately implemented verifier. The human orchestrator has not independently checked every tensor calculation; the manuscript therefore remains subject to expert review.

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1 Question, contribution, and claim boundary

For a stationary black hole in an asymptotically flat theory, the normalization of the horizon generator is usually fixed by the asymptotic clock. Generic static spherical solutions of pure Weyl gravity are not asymptotically flat, and their Schwarzschild-gauge representatives retain a local radial-diffeomorphism/Weyl freedom. The chart vector ∂_t is consequently not a residual-basic object.

This matters already on the three-dimensional static parameter family. The Iyer–Wald surface form for ∂_t is radially conserved but not integrable in parameter space. Adding a parameter-local corner function cannot cure this, because it cannot change the exterior derivative of the charge one-form. Our central observation is that the obstruction disappears when the generator is normalized as an object on the residual quotient. Basicness determines the normalization up to a function of the sole quotient invariant, and the resulting charge is closed.

The static Bach-flat family itself is classical [2, 3]; conformal-gravity black-hole thermodynamics and additional spin-two hair have also been studied in asymptotically AdS and Einstein–Weyl settings [9]. The contribution here is not a new solution family. It is:

1. an exact residual-quotient derivation of the field-dependent normalization;
2. an exact residual-basic Hamiltonian and Wald entropy in that normalization;
3. one first-law identity holding simultaneously at every simple horizon;
4. a linear spherical audit showing that the charge is not selected by a conformal frame or by a spherical coordinate gauge.

Lü, Pang, Pope, and Vázquez-Poritz use an AdS ensemble with an independent thermodynamic pair for massive spin-two hair [9]. Our one-term identity is instead a quotient statement on a restricted static family with no preferred asymptotic clock. The two constructions therefore answer different normalization and ensemble questions.

The phrase “first law” below means an exact variational identity on this finite-dimensional static solution family. It does not assert that all roots are simultaneous equilibrium boundaries of one thermal ensemble, and it does not identify H with a preferred asymptotic mass.

1.1 Dependency closure

Table 1 separates the exact theorem chain from open extensions. The dependency tags are those used in the computational archive. In the compact status column, LA means LOCAL-ALGEBRAIC and RM means REDUCED-MODE; the abbreviated certificate names are expanded in Section 10.

Result	Evidence	Status	Role in main theorem
Laurent Bach-flat locus, residual action, horizon fixture	BH0	LA, exact	Background and quotient geometry
Bare static Iyer–Wald form	BH1	LA+RM, exact	Exhibits the normalization obstruction
Basic generator, H, S, T , simultaneous first law	BH1A, P18	LA+RM, exact	Main static theorem
Spherical conformal/diffeomorphism audit	BH1B	LA+RM, exact at linear charge level	Frame and gauge independence
Radiative flux, physical-process law, stability	none	open	Not used and not claimed

Table 1: Logical dependency of the paper. “Exact” means symbolic rational or polynomial identity, not floating-point evidence.

2 Conventions and the static Laurent family

We use signature $(-+++)$ and action

$$S_W[g] = \alpha \int_M \sqrt{-g} C_{abcd} C^{abcd} d^4x. \quad (1)$$

Our Bach-tensor convention is

$$B_{ab} = \nabla^c \nabla^d C_{acbd} + \frac{1}{2} R^{cd} C_{acbd}. \quad (2)$$

The field equation is $B_{ab} = 0$.

Start from the areal-radius ansatz

$$ds^2 = -a(r)dt^2 + b(r)dr^2 + r^2 d\Omega_2^2. \quad (3)$$

Before conformal gauge fixing, the exact Bach tensor is diagonal, trace-free, divergence-free, and has two independent components. We then choose $b = 1/a$, write $a = B$, and restrict to

$$B(r) = w - \frac{u}{r} + \gamma r - kr^2 + \frac{c_2}{r^2} + c_3 r^3. \quad (4)$$

Theorem 2.1 (Bach-flat locus in the declared Laurent class). *The metric (4) is Bach-flat if and only if*

$$c_2 = c_3 = 0, \quad w^2 + 3u\gamma = 1. \quad (5)$$

Proof. Substitution of (4) into the two independent Bach rows, clearing the declared denominators and equating Laurent coefficients, gives an exact polynomial ideal. Its reduced Gröbner basis is

$$\{c_2, c_3, w^2 + 3u\gamma - 1\}.$$

This proves necessity and sufficiency within the ansatz. The producer and independent curvature implementation both recompute the vanishing tensor; mutating the correlated constant term or adding an r^{-2} tail produces explicit nonzero Bach components. $\square$

The two reduced Bach rows and the exact coefficient-ideal construction are displayed in Section 9. They make the Laurent classification reconstructible from the manuscript rather than only from the machine-readable certificate.

Remark 2.2 (Scope of completeness). Theorem 2.1 is not presented as a new proof of the full Birkhoff–Riegert theorem [2]. Generalized Birkhoff statements and nonstandard spherical branches require additional global and nondegeneracy hypotheses [10]. Here “complete” always means complete in (4).

The Mannheim–Kazanas chart is

$$B(r) = 1 - 3\beta\gamma - \frac{\beta(2 - 3\beta\gamma)}{r} + \gamma r - kr^2, \quad (6)$$

so

$$u = \beta(2 - 3\beta\gamma), \quad w = 1 - 3\beta\gamma. \quad (7)$$

It parametrizes the component through $w = 1$ used in the charge theorem.

Proposition 2.3 (Correct Einstein locus). *On the full Laurent locus (5), the metric is Einstein in the areal-radius Schwarzschild chart if and only if*

$$\gamma = 0, \quad w = 1. \quad (8)$$

On the Mannheim–Kazanas component (6), this reduces to $\gamma = 0$.

Proof. On the full locus the scalar curvature is

$$R = 12k - \frac{6\gamma}{r} + \frac{2(1 - w)}{r^2}. \quad (9)$$

An Einstein metric has constant scalar curvature, forcing $\gamma = 0$ and $w = 1$. Direct substitution then gives $R_{ab} = 3kg_{ab}$. In the MK chart, the trace-free Ricci defect contains

$$E_{\theta\theta} = -\frac{\gamma}{2}(r - 3\beta),$$

so $\gamma = 0$ is necessary and sufficient there. The allowed Laurent point $(\gamma, w) = (0, -1)$ is therefore Bach-flat but not Einstein; it belongs to a complementary Kantowski–Sachs-type chart rather than to the MK component through $w = 1$. $\square$

3 Residual quotient and its invariant

Put $x = 1/r$ and

$$Q(x) = -ux^3 + wx^2 + \gamma x - k. \quad (10)$$

The roots of Q encode the nonzero roots of B . The form-preserving local transformation

$$r = \frac{\rho}{1 - c\rho}, \quad g \mapsto (1 - c\rho)^2 g \quad (11)$$

acts as the translation $Q(x) \mapsto Q(x - c)$. Equivalently,

$$\begin{aligned} u &\mapsto u, & w &\mapsto w + 3cu, \\ \gamma &\mapsto \gamma - 2cw - 3c^2u, & k &\mapsto k + c\gamma - c^2w - c^3u. \end{aligned} \quad (12)$$

A dilation acts by

$$(\beta, \gamma, k) \mapsto (\beta/\lambda, \lambda\gamma, \lambda^2k), \quad Q(x) \mapsto \lambda^2 Q(x/\lambda). \quad (13)$$

The conformal factor in the c -map may vanish at finite radius, so (12) is used as a local quotient statement, not as a global identification of black-hole exteriors.

On the MK parameters the infinitesimal generators are

$$X_c = -3\beta^2\partial_\beta + (6\beta\gamma - 2)\partial_\gamma + \gamma\partial_k, \quad (14)$$

$$X_\lambda = -\beta\partial_\beta + \gamma\partial_\gamma + 2k\partial_k. \quad (15)$$

They have generic rank two. A continuous invariant is

$$J = u^2 \text{disc}(Q). \quad (16)$$

Translation invariance of the discriminant and its scaling weight prove $X_c J = X_\lambda J = 0$. Moreover, $\text{disc}(Q) = 0$ is precisely the multiple-root, hence degenerate-horizon, locus. On an open set where the orbit rank is two and $dJ \neq 0$, J is a local coordinate on the one-dimensional quotient. Every smooth local invariant is therefore a function of J . This statement neither identifies distinct connected components nor follows a labelled horizon through a multiple-root collision.

4 The bare charge is not integrable

For a diffeomorphism-covariant Lagrangian depending algebraically on the Riemann tensor, define

$$E^{abcd} = \frac{\partial(\alpha C_{efgh} C^{efgh})}{\partial R_{abcd}} = 2\alpha C^{abcd}.$$

We use the Iyer–Wald representative [4, 5]

$$\theta^a(g; \delta g) = 2 \left(E^{abcd} \nabla_d \delta g_{bc} - \delta g_{bc} \nabla_d E^{abcd} \right), \quad (17)$$

$$Q_\xi^{ab} = -E^{abcd} \nabla_c \xi_d + 2\xi_d \nabla_c E^{abcd}. \quad (18)$$

The sphere conventions are

$$(Q_\xi)_{\theta\phi} = 2\sqrt{-g} Q_\xi^{tr}, \quad (i_\xi \theta)_{\theta\phi} = -\sqrt{-g} \xi^t \theta^r.$$

With $\xi = \partial_t$, integration over any sphere gives the parameter-space one-form

$$\mathcal{F} = F_\beta d\beta + F_\gamma d\gamma + F_k dk, \quad (19)$$

where

$$\begin{aligned} F_\beta &= 16\pi\alpha(12\beta\gamma k - \gamma^2 - 4k), \\ F_\gamma &= 16\pi\alpha\beta(6\beta k - \gamma), \\ F_k &= -16\pi\alpha\beta(2 - 3\beta\gamma). \end{aligned} \quad (20)$$

Each component is exactly independent of the sphere radius. As a normalization control, replacing E^{abcd} by the Einstein–Hilbert tensor reproduces $16\pi \delta m$ on Schwarzschild and Schwarzschild–de Sitter in the convention $16\pi G = 1$.

Proposition 4.1 (Bare nonintegrability). *The chart-normalized form $\mathcal{F}$ is not closed:*

$$d\mathcal{F} = 16\pi\alpha [\gamma d\beta \wedge d\gamma + 2(1 - 3\beta\gamma) d\beta \wedge dk - 3\beta^2 d\gamma \wedge dk]. \quad (21)$$

It obeys

$$i_{X_c} \mathcal{F} = 0, \quad i_{X_c} d\mathcal{F} = 0, \quad (22)$$

$$i_{X_\lambda} \mathcal{F} = 0, \quad i_{X_\lambda} d\mathcal{F} = \mathcal{F}. \quad (23)$$

Consequently no parameter-local boundary function can make $\mathcal{F}$ integrable.

Proof. Equations (21)–(23) follow by exact differentiation of (20). If a boundary function W were added, the one-form would change by dW , leaving $d\mathcal{F}$ unchanged. The kernel of $d\mathcal{F}$ is the X_c direction at generic points. Thus quotienting only the translation leaves a nondegenerate two-form on the remaining two-dimensional parameter quotient; the obstruction is not removed merely by discarding that proper-gauge direction. $\square$

5 Basicness fixes the field-dependent generator

For a field-dependent generator $\chi = N(\beta, \gamma, k)\partial_t$, we use the corrected surface form

$$k_\chi(g; \delta g) = \delta Q_\chi - Q_{\delta\chi} - i_\chi \theta(g; \delta g). \quad (24)$$

Because N is spacetime-constant on each static solution and the Noether charge is linear in χ , the $Q_{\delta\chi}$ term cancels all dN contributions. The corrected charge is therefore

$$\delta H_\chi = N\mathcal{F}. \quad (25)$$

Here the variation acts on both the dynamical fields and the field-dependent vector $\chi[g]$; the explicit $Q_{\delta\chi}$ subtraction removes the latter contribution. This is the field-dependent-generator extension of the Iyer–Wald form used in field-dependent asymptotic symmetry algebras [8, 7]. We fix the Lagrangian (1), the displayed θ , and the displayed Q_χ . The standard Jacobson–Kang–Myers shifts of the Lagrangian or presymplectic potential can change a surface representative by boundary terms [6]; no such shift is made here. The result is therefore a theorem about this explicit representative on closed symmetry spheres, not a uniqueness assertion over every covariant-phase-space ambiguity.

Definition 5.1. A parameter-space form is *residual-basic* if it is horizontal and invariant under both vector fields (14) and (15).

Theorem 5.2 (Residual-basic normalization). *Let $\mathcal{U}$ be a connected open subset of the MK parameter space on which $u \neq 0$, the residual orbit rank is two, $dJ \neq 0$, and $\mathcal{F} \neq 0$. If N is a smooth nowhere-zero normalization on $\mathcal{U}$, then the corrected charge $N\mathcal{F}$ is residual-basic only if*

$$N = uf(J) \quad (26)$$

for a function f of the quotient invariant. Conversely, every smooth normalization (26) that remains nowhere zero on $\mathcal{U}$ is basic and integrable. For $f = 1$,

$$u\mathcal{F} = dH, \quad H = -16\pi\alpha\beta^2 D_2, \quad (27)$$

where

$$D_2 = 9\beta^2\gamma^2 k - \beta\gamma^3 - 12\beta\gamma k + \gamma^2 + 4k. \quad (28)$$

Proof. Cartan's identity and (22) give $\mathcal{L}_{X_c}\mathcal{F} = 0$. Hence invariance of $N\mathcal{F}$ requires $X_c N = 0$. Similarly, (23) gives $\mathcal{L}_{X_\lambda}\mathcal{F} = \mathcal{F}$, so invariance requires $X_\lambda N = -N$. Directly,

$$X_c u = 0, \quad X_\lambda u = -u.$$

Since the residual orbit has codimension one and J is a local quotient coordinate on $\mathcal{U}$, the general local solution is $N = uf(J)$.

Exact differentiation of (27) gives $dH = u\mathcal{F}$, proving closure for $f = 1$. The same form is horizontal under both residual fields. For general $f(J)$, $f(J)dH$ is closed because the exact identity $dH \wedge dJ = 0$ holds. It is therefore locally the differential of a reparametrized quotient Hamiltonian. $\square$

Remark 5.3 (Local uniqueness and exceptional loci). The hypotheses in Theorem 5.2 are essential. At a zero of $\mathcal{F}$, invariance of $N\mathcal{F}$ alone need not determine N ; where $dJ = 0$, J is not a quotient coordinate; and at $u = 0$ the canonical normalization degenerates. A normalization obtained on a regular component may extend continuously to such a locus, but that extension is not another local uniqueness theorem. Discrete component data and permutations of horizon roots are likewise outside the local classification.

The discriminant factorization sharpens the quotient interpretation. Put

$$D_1 = 27\beta^2 k - 3\beta\gamma - 1. \quad (29)$$

Then

$$J = -u^2 D_1 D_2. \quad (30)$$

The Hamiltonian vanishes on the $D_2 = 0$ component of the degenerate-horizon locus. On the Einstein control family $\gamma = 0$,

$$H = -64\pi\alpha\beta^2 k. \quad (31)$$

On the Schwarzschild subfamily $k = \gamma = 0$, it vanishes identically.

Remark 5.4 (Normalization versus physical mass). Equation (27) defines the residual-basic Hamiltonian of the generator on the declared static phase-space slice. We do not call it a universal physical mass. Multiplication by $f(J)$, component orientation, and any future choice of physical asymptotic clock change the normalization. On a fixed-falloff ensemble with γ and k fixed, the nontrivial residual transformations are generically frozen, but this does not create a preferred clock on the full family.

6 Entropy and simultaneous first laws

Let r_h be any simple positive root of B . The Wald entropy is

$$S_h = -2\pi \int_{S_{r_h}} E^{abcd} \epsilon_{ab} \epsilon_{cd} dA. \quad (32)$$

With $E^{abcd} = 2\alpha C^{abcd}$ and $\epsilon_{tr} = 1$, exact evaluation gives

$$S_h = 64\pi^2 \alpha \beta \frac{2 - 3\beta\gamma + \gamma r_h}{r_h}. \quad (33)$$

For the algebraic orientation $\chi = u\partial_t$, define the signed horizon temperature

$$T_h = \frac{\kappa_\chi}{2\pi} = \frac{uB'(r_h)}{4\pi}. \quad (34)$$

This is the signed surface-gravity quantity entering the oriented variational identity; a positive Hawking temperature would use $|uB'(r_h)|/(4\pi)$. We do not infer a common thermal-equilibrium interpretation when several simple roots are present.

Theorem 6.1 (Simultaneous horizon first laws). *For every locally labelled simple root r_h of (6), the residual-basic Hamiltonian (27), entropy (33), and signed horizon temperature (34) satisfy*

$$\boxed{dH = T_h dS_h} \quad (35)$$

in all three parameter directions. If one solution has simple roots $r_1, \dots, r_n$, then

$$T_1 dS_1 = \dots = T_n dS_n = dH. \quad (36)$$

Proof. Along the horizon locus $B(r_h) = 0$, any parameter $p \in \{\beta, \gamma, k\}$ obeys

$$\frac{dr_h}{dp} = -\frac{\partial_p B(r_h)}{B'(r_h)}. \quad (37)$$

Use this in the total derivative of (33), subtract it from $\partial_p H/T_h$, clear denominators, and reduce the numerator modulo

$$r_h B(r_h) = 0. \quad (38)$$

The three exact remainders are displayed in Section 9. They are proportional to $B(r_h)$, or vanish before imposing the horizon equation. The calculation uses no choice of root beyond local simplicity, proving the result at all locally labelled roots simultaneously. $\square$

For Schwarzschild, $r_h = 2\beta$, so

$$S_h = 64\pi^2 \alpha$$

is independent of β , consistently with $H = 0$ on that subensemble. No pressure–volume or extra source term is needed for (35) on the declared residual-basic static family. This does not exclude such terms in a different ensemble with independently varied boundary data.

6.1 Exact non-Einstein three-horizon fixture

Consider

$$(\beta, \gamma, k) = \left(\frac{3}{2}, \frac{12}{19}, \frac{1}{19} \right). \quad (39)$$

Then

$$rB(r) = -\frac{1}{19}(r-1)(r-3)(r-8), \quad u = -\frac{24}{19}, \quad (40)$$

and the trace-free Ricci witness is

$$E_{\theta\theta} = -\frac{3(2r-9)}{19} \neq 0.$$

The metric is smooth at all three roots in ingoing Eddington–Finkelstein coordinates, and all certified scalar curvature invariants are finite there. In the algebraic $N = u$ orientation,

$$H = -\frac{14400\pi\alpha}{6859} \quad (41)$$

and

r_h	S_h	T_h
1	$-384\pi^2\alpha/19$	$84/(361\pi)$
3	$640\pi^2\alpha/19$	$-20/(361\pi)$
8	$960\pi^2\alpha/19$	$105/(1444\pi)$

The three identities $T_h dS_h = dH$ hold exactly in every parameter direction. Since $u < 0$ on this component, choosing the opposite future-directed orientation flips H and every signed T_h , leaving the variational identities unchanged.

7 Linear spherical frame and gauge audit

The preceding derivation lives on the static parameter slice. It is nevertheless important to decide whether its normalization secretly selects a conformal frame. We therefore consider the complete linear spherical gauge sector around both symbolic Schwarzschild and the non-Einstein fixture:

$$\delta_\omega g_{ab} = 2\omega(t, r)g_{ab}, \quad (42)$$

$$\delta_\xi g_{ab} = \mathcal{L}_\xi g_{ab}, \quad \xi = a(t, r)\partial_t + b(t, r)\partial_r. \quad (43)$$

Theorem 7.1 (Linear spherical gauge independence). *At linear charge level:*

1. $k_\chi(2\omega g) = 0$ componentwise for arbitrary $\omega(t, r)$;
2. the entropy variation under $2\omega g$ vanishes on the symbolic MK family;
3. the corrected presymplectic current between a conformal variation and any static parameter mode vanishes;
4. $k_\chi(\mathcal{L}_\xi g) = 0$ for arbitrary spherical ξ ;
5. the extension of $N = u$ is uniquely fixed by $\delta N = du$ on parameter directions and $\delta N = 0$ on spherical conformal and diffeomorphism directions.

Thus no boundary clock or preferred conformal representative is needed for the static theorem at this linear charge level.

Proof. The conformal statements follow from direct substitution in (24), exact conformal covariance of the Bach equation, and the literal variation of the density $\sqrt{-g}\theta^a$. The density variation is essential: omitting the trace contribution spoils current conservation. The diffeomorphism statement is verified through the on-shell Noether identity

$$\theta(\mathcal{L}_\xi g) - i_\xi(L\epsilon) = dQ_\xi \quad (44)$$

and the background identity obtained from its commutator with χ . Both routes are independently evaluated in the certificate.

For uniqueness, the bare Noether aperture is nonzero on the fixture. Any nonzero δN on a certified gauge direction would therefore assign that direction a nonzero charge, contradicting the componentwise annihilation above. The parameter variation of N is already fixed by Theorem 5.2. $\square$

The theorem is complete for the $\ell = 0$ dynamical gauge sector at linear charge level. It does not compute the bilinear radiative flux for $\ell \geq 2$, and it does not imply a second-order physical-process first law.

8 Ensembles, interpretation, and non-results

The c -transformation preserves the fixed-falloff data $\{\gamma, k\}$ only for $c = 0$, away from the algebraic locus

$$u(w^2 - 3) = 0.$$

The $u = 0$ component is excluded because the basic normalization degenerates. The dilation preserves the same ensemble only at $\lambda = 1$, except on the Schwarzschild subensemble $\gamma = k = 0$, where it acts freely and consistently with $H = 0$ and constant entropy. On a finite exterior $[r_h, r_{\text{out}}]$, the c -map conformal factor is smooth and positive precisely when $1 + cr > 0$ throughout that interval.

These observations support a precise interpretation:

The nonintegrability of the chart-normalized charge is a normalization artifact. Requiring the generator and charge to descend to the complete local residual quotient selects an integrable class of generators. It does not, by itself, choose a unique physical clock.

The following statements are not established:

- completeness of static Bach vacua beyond the Laurent ansatz;
- a preferred physical matter or clock conformal frame;
- a universal physical mass valid across inequivalent boundary ensembles;
- a nonlinear or second-order physical-process first law;
- the bilinear radiative symplectic flux through a horizon or infinity;
- stability, quasinormal ringing, Hawking radiation, or any quantum property.

9 Exact audit algebra and reconstructible proofs

This section exposes the shortest load-bearing symbolic identities used in the theorem. Write $B^{(j)} = d^j B / dr^j$. In the gauge $a = B$, $b = B^{-1}$, two independent Bach rows are

$$B_{tt} = \frac{B}{24r^4} \mathcal{R}_t, \quad B_{rr} = \frac{1}{24r^4 B} \mathcal{R}_r, \quad (45)$$

where

$$\begin{aligned} \mathcal{R}_t = & 4r^4 B B^{(4)} + 2r^4 B' B''' - r^4 (B'')^2 + 12r^3 B B''' + 4r^3 B' B'' \\ & - 4r^2 B B'' - 4r^2 (B')^2 + 8r B B' - 4B^2 + 4, \end{aligned} \quad (46)$$

$$\begin{aligned} \mathcal{R}_r = & -2r^4 B' B''' + r^4 (B'')^2 + 4r^3 B B''' - 4r^3 B' B'' \\ & + 4r^2 B B'' + 4r^2 (B')^2 - 8r B B' + 4B^2 - 4. \end{aligned} \quad (47)$$

The angular row follows from tracelessness. Substituting (4) into (46)–(47), clearing powers of r , and collecting every coefficient generates the ideal

$$I_{\text{Laurent}} = \langle [r^j] \mathcal{R}_t, [r^j] \mathcal{R}_r \mid j \in \mathbb{Z} \rangle \subset \mathbb{Q}[c_2, c_3, u, w, \gamma, k].$$

With lexicographic order $c_2 > c_3 > u > w > \gamma > k$, its reduced Gröbner basis is

$$\text{GB}(I_{\text{Laurent}}) = \{c_2, c_3, w^2 + 3u\gamma - 1\}. \quad (48)$$

This is the finite algebraic input to Theorem 2.1; the complete coefficient list is retained in the BH0 producer rather than typeset as 48 redundant multiples.

The first-law reduction is shorter still. Scale out the universal constants by putting

$$h = \frac{H}{16\pi\alpha} = -\beta^2 D_2, \quad s = \frac{S_h}{64\pi^2\alpha} = \frac{\beta(2 - 3\beta\gamma + \gamma r_h)}{r_h}. \quad (49)$$

For $p \in \{\beta, \gamma, k\}$, define

$$\mathcal{E}_p = \partial_p h - u [B'(r_h) \partial_p s - (\partial_{r_h} s) \partial_p B(r_h)]. \quad (50)$$

This is exactly $\partial_p H - T_h dS_h/dp$, divided by $16\pi\alpha$, after implicit differentiation of the horizon equation. Direct reduction gives the rational identities

$$\mathcal{E}_\beta = B(r_h) \frac{2\beta\gamma(3\beta\gamma - 2)}{r_h}, \quad (51)$$

$$\mathcal{E}_\gamma = B(r_h) \frac{2\beta^2(3\beta\gamma - 2)}{r_h}, \quad (52)$$

$$\mathcal{E}_k = 0. \quad (53)$$

They prove the three parameter components of (35) immediately on $B(r_h) = 0$. The standard-library verifier cited below represents these expressions as sparse Laurent polynomials with exact rational coefficients and checks them coefficientwise without SymPy.

10 Computational evidence and reproduction

The calculations are computer-assisted but exact. Tensor components, polynomial reductions, and charge identities are represented by symbolic rational functions. No theorem in this paper is inferred from a decimal approximation. Programme producer and verifier use separately written curvature pipelines; the verifier constructs the Weyl tensor through an independent Schouten/Kulkarni–Nomizu route. A further Paper 18 rail uses only the Python standard library and an independently implemented sparse Laurent-polynomial ring over `fractions.Fraction`. Mutation tests ensure that certificate-like but false inputs are rejected.

Certificate	Reproduction and independent verification
BH0_STATIC_SPHERICAL_BACKGROUND.json	python3 black_hole_programme/bh0.background.py python3 black_hole_programme/verify_bh0.background.py
BH1_LEE_WALD_PREFLIGHT.json	python3 black_hole_programme/bh1_lee_wald_preflight.py python3 black_hole_programme/verify_bh1_lee_wald_preflight.py
BH1A_NORMALIZED_GENERATOR.json	python3 black_hole_programme/bh1a_normalized_generator.py python3 black_hole_programme/verify_bh1a_normalized_generator.py
BH1B_DYNAMICAL_EXTENSION.json	python3 black_hole_programme/bh1b_dynamical.py python3 black_hole_programme/verify_bh1b_dynamical.py
PAPER18_STATIC_FIRST_LAW_PROMOTION.json	python3 black_hole_programme/paper18_static_first_law_promotion.py python3 black_hole_programme/verify_paper18_static_first_law_promotion.py
Independent algebra rail	python3 paper/verify_18_static_weyl_thermodynamics_stdlib.py

Table 2: Authoritative certificate families. Re-running a producer is reproduction; the separately implemented verifier is the independent check.

The paper-specific promotion certificate is append-only. It leaves the historical BH1, BH1A, and BH1B PREFLIGHT lifecycle labels untouched and records a narrower CLASSIFIED successor for exactly the static MK theorem and linear spherical charge audit stated here. It does not promote a physical-process law, a radiative flux law, or a peer-review status.

The BH0 mutations (i) remove the correlated $-3\beta\gamma$ constant and (ii) add an r^{-2} tail to the three-horizon fixture; both yield nonzero Bach components. The bare-charge computation includes Einstein normalization controls and retains $d\mathcal{F} \neq 0$ as a negative control. The BH1A verifier independently recomputes entropy and reduces the first-law polynomial at all three fixture roots. The BH1B verifier recomputes the conformal and diffeomorphism annihilation identities.

11 Conclusion

Static Weyl gravity does not supply a preferred Killing normalization merely by presenting a Schwarzschild-gauge metric. On the certified Mannheim–Kazanas family, treating ∂_t as fixed produces a radially conserved but nonintegrable Iyer–Wald form. Treating the generator as field dependent and demanding residual basicness instead forces $\chi = uf(J)\partial_t$. Its charge is integrable,

its Wald entropy is explicit, and one residual-basic Hamiltonian variation satisfies a signed horizon first-law identity at every locally labelled simple horizon simultaneously.

The result is stronger than a coordinate identity and weaker than a complete black-hole thermodynamics. It is invariant under the complete linear spherical conformal and diffeomorphism gauge sector, but it does not fix a physical asymptotic clock or control nonlinear radiative flux. Those boundaries are part of the theorem, not deferred qualifications.

References

- [1] R. Bach, *Zur Weylschen Relativitätstheorie und der Weylschen Erweiterung des Krümmungstensorbegriffs*, Math. Z. **9** (1921) 110–135.
- [2] R. J. Riegert, *Birkhoff’s theorem in conformal gravity*, Phys. Rev. Lett. **53** (1984) 315–318.
- [3] P. D. Mannheim and D. Kazanas, *Exact vacuum solution to conformal Weyl gravity and galactic rotation curves*, Astrophys. J. **342** (1989) 635–638.
- [4] J. Lee and R. M. Wald, *Local symmetries and constraints*, J. Math. Phys. **31** (1990) 725–743.
- [5] V. Iyer and R. M. Wald, *Some properties of Noether charge and a proposal for dynamical black hole entropy*, Phys. Rev. D **50** (1994) 846–864, [arXiv:gr-qc/9403028](#).
- [6] T. Jacobson, G. Kang, and R. C. Myers, *On black hole entropy*, Phys. Rev. D **49** (1994) 6587–6598, [arXiv:gr-qc/9312023](#).
- [7] R. M. Wald and A. Zoupas, *A general definition of “conserved quantities” in general relativity and other theories of gravity*, Phys. Rev. D **61** (2000) 084027, [arXiv:gr-qc/9911095](#).
- [8] G. Barnich and C. Troessaert, *BMS charge algebra*, JHEP **12** (2011) 105, [arXiv:1106.0213](#).
- [9] H. Lü, Y. Pang, C. N. Pope, and J. F. Vázquez-Poritz, *AdS and Lifshitz black holes in conformal and Einstein–Weyl gravities*, Phys. Rev. D **86** (2012) 044011, [arXiv:1204.1062](#).
- [10] P. Jizba and T. Lehečková, *Generalized Birkhoff theorems and $2 + 2$ direct product spacetimes in Weyl conformal gravity*, Phys. Rev. D **113** (2026) 044060, [doi:10.1103/9g8r-lrjk](#), [arXiv:2512.18482](#).